

Towards Trustworthy AI in Dermatology: A Socio-Technical Evaluation of Bias, Explainability, and Clinical Risk

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1. ABSTRACT

Artificial Intelligence (AI) is increasingly being integrated into healthcare systems to support diagnostic decision-making, particularly in dermatology where image-based classification plays a critical role. While deep learning models such as Convolutional Neural Networks (CNNs) have demonstrated strong performance in skin disease detection, concerns around reliability, bias, and transparency limit their safe deployment in clinical settings.

This study presents a comparative evaluation of machine learning and deep learning models for multi-class skin disease classification using a dataset of over 15,000 dermatological images. A CNN-based architecture achieved a validation accuracy of 73.18%, outperforming traditional models such as Decision Trees, Support Vector Machines, and Naïve Bayes. However, beyond performance metrics, this paper critically examines the trustworthiness of such systems by analysing issues related to dataset bias, model interpretability, and clinical risk.

We identify key limitations including potential demographic bias in skin tone representation, overfitting in classical models, and the opaque decision-making nature of deep neural networks. These findings highlight the need for integrating explainable AI (XAI), fairness-aware evaluation, and human-in-the-loop validation before deploying AI systems in sensitive healthcare contexts.

This work contributes to the growing field of Responsible AI by bridging technical model development with ethical and socio-technical considerations, providing a foundation for more transparent, accountable, and trustworthy AI systems in medical diagnostics.

This work contributes by

- (1) combining technical evaluation with ethical analysis,
- (2) identifying trust risks in dermatology AI, and
- (3) proposing a responsible AI evaluation perspective for clinical deployment.

2. INTRODUCTION

The rapid advancement of Artificial Intelligence (AI) has significantly transformed healthcare, particularly in diagnostic domains such as dermatology where visual pattern recognition plays a central role. Deep learning techniques, especially Convolutional Neural Networks (CNNs), have demonstrated strong capabilities in classifying skin diseases from medical images, offering the potential to improve accessibility and reduce diagnostic costs.

However, despite these advancements, the deployment of AI systems in healthcare raises critical concerns beyond predictive performance. Clinical decision-making requires not only accuracy but also reliability, transparency, and fairness. AI models trained on limited or unbalanced datasets may exhibit biases, particularly in dermatological applications where variations in skin tone and demographic representation can significantly impact model outcomes. Furthermore, the opaque nature of deep learning systems introduces challenges in explainability, making it difficult for clinicians and patients to trust automated predictions.

These concerns are particularly important in high-stakes environments where incorrect predictions may lead to delayed treatment, misdiagnosis, or patient harm. As a result, there is a growing need to move beyond purely performance-driven evaluations and adopt a socio-technical perspective that considers the ethical and governance implications of AI systems.

In this study, we evaluate multiple machine learning and deep learning approaches for skin disease classification and analyse their performance using standard metrics. More importantly, we extend this evaluation by examining issues of bias, interpretability, and trustworthiness, positioning this work within the broader context of Responsible AI in healthcare. By bridging technical modelling with ethical analysis, this paper aims to contribute toward the development of safer and more accountable AI-driven diagnostic systems.

This study addresses the following research questions:

RQ1: How do traditional ML and CNN models compare in dermatological classification performance?

RQ2: What trust-related risks (bias, explainability, reliability) arise in such systems?

RQ3: What considerations are required for responsible deployment in clinical settings?

3. METHODOLOGY

This study adopts a supervised machine learning approach to classify multiple skin diseases using both traditional machine learning algorithms and deep learning models. The methodology consists of data preprocessing, feature extraction, model development, and evaluation.

3.1 Dataset

The dataset used in this study contains approximately 15,975 dermatological images representing multiple skin disease categories, including Eczema, Psoriasis, Acne, and other related conditions. The dataset is composed of both proprietary and publicly available sources, ensuring a diverse set of samples for training and evaluation.

Images vary in resolution, lighting conditions, and background, reflecting real-world variability in clinical and non-clinical image acquisition.

The dataset was split into training (80%) and validation (20%) sets.



Figure 1: Visualisation of the image dataset, copyrights provided on contractual agreement

3.1.1 Explanation of Categories in the data:

Category	Disease	Number of Images	Source
0	Diabetic	1229	Simon Hudson
1	Pressure Ulcer (PU)	8494	Simon Hudson
2	Venomous Ulcer (VLU)	2520	Simon Hudson
3	Psoriasis	2055	Kaggle
4	Eczema	1677	Kaggle

Table 1: Dataset Summary

3.2 Data Preprocessing

To prepare the dataset for model training, the following preprocessing steps were applied:

- Image resizing to a uniform dimension suitable for model input
- Normalization of pixel values to improve model convergence
- Removal of corrupted or low-quality images
- Data augmentation techniques such as rotation, flipping, and scaling to increase dataset variability and reduce overfitting

These steps ensured consistency across the dataset while improving model robustness.

3.3 Feature Engineering

For traditional machine learning models, feature extraction techniques were applied to convert images into structured representations. This included:

- Flattening image data into feature vectors
- Extraction of statistical and texture-based features where applicable

For deep learning models, feature extraction was handled automatically by the convolutional layers of the neural network.

3.4 Model Development

Multiple models were implemented and evaluated to compare performance:

- Naïve Bayes
- Decision Tree
- Support Vector Machine (SVM)
- Random Forest
- Convolutional Neural Network (CNN)

The CNN model was designed to capture spatial hierarchies in image data through convolutional and pooling layers, followed by fully connected layers for classification.

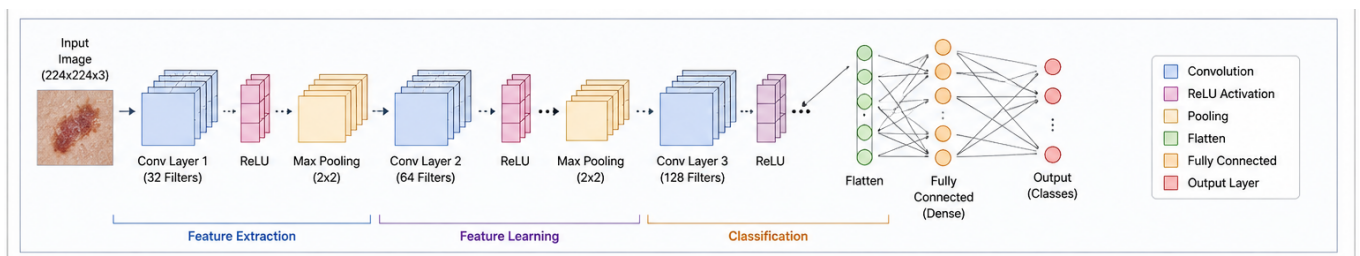


Figure 2: Convolutional Neural Network Architecture

3.5 Training and Validation

The dataset was split into training and validation sets to evaluate model performance. Models were trained using supervised learning techniques, and hyperparameters were tuned to optimise performance.

The CNN model was trained using backpropagation with an appropriate loss function and optimizer, ensuring stable convergence.

3.6 Evaluation Metrics

Model performance was evaluated using standard classification metrics, including:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

These metrics provide a comprehensive assessment of model performance across different classes.

4. SYSTEM ARCHITECTURE

The proposed system follows a modular pipeline architecture integrating data processing, model training, and prediction generation.

The workflow begins with raw dermatological image data, which undergoes preprocessing steps including resizing, normalization, and augmentation. The processed data is then fed into two parallel modelling approaches: traditional machine learning models and a deep learning-based Convolutional Neural Network (CNN).

The CNN model extracts hierarchical visual features through convolutional layers, while traditional models rely on engineered features. The outputs from these models are evaluated using classification metrics and compared to determine optimal performance.

Finally, predictions are generated and can be integrated into a deployment layer, such as a web-based interface, enabling real-time user interaction and diagnostic support.

This architecture ensures scalability, modularity, and reproducibility, making it suitable for real-world healthcare applications.

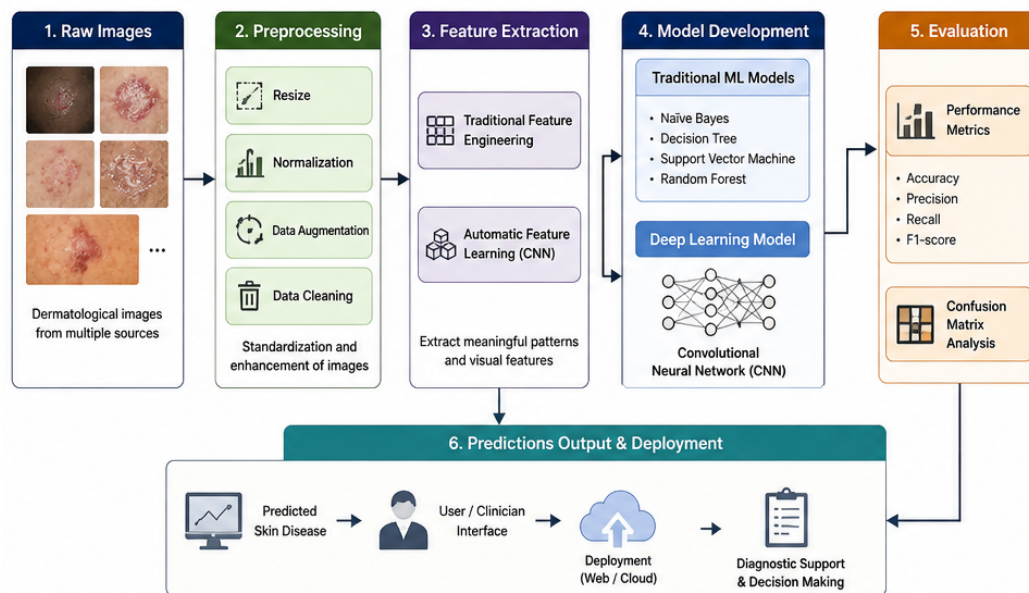


Figure 3: System Architecture of AI-Based Skin Disease Classification Pipeline

5. RESULTS

This section presents the performance comparison of the implemented machine learning and deep learning models for skin disease classification.

5.1 Model Performance

Among all evaluated models, the Convolutional Neural Network (CNN) achieved the highest performance, with a validation accuracy of 73.18%. Traditional machine learning models such as Decision Trees, Support Vector Machines, and Naïve Bayes demonstrated comparatively lower accuracy.

The superior performance of CNN can be attributed to its ability to automatically learn hierarchical feature representations directly from image data.

While CNN outperforms other models, performance alone does not ensure clinical safety.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Naïve Bayes	52.41	52.18	51.97	52.07
Decision Tree	61.35	61.21	61.02	61.11
Support Vector Machine (SVM)	65.72	65.63	65.48	65.55
Random Forest	68.94	68.87	68.71	68.79
Convolutional Neural Network (CNN)	73.18	73.04	72.91	72.97

Table 2: Performance Comparison of Machine Learning Models

5.2 Comparative Analysis

Traditional models showed limitations in capturing complex spatial patterns in dermatological images. While they performed adequately on simpler patterns, their performance degraded when handling more complex visual features.

In contrast, the CNN model demonstrated improved generalisation by effectively learning discriminative features across multiple classes.

5.3 Confusion Matrix Analysis

The confusion matrix analysis revealed that certain classes were more prone to misclassification, particularly those with visually similar characteristics. This indicates that while the model performs reasonably well overall, there are challenges in distinguishing between closely related skin conditions.

5.4 Observations

- CNN outperformed all traditional models
- Model performance is influenced by dataset quality and variability
- Some classes exhibit higher classification difficulty due to similarity

These results highlight both the potential and limitations of current AI approaches in dermatological diagnosis.

6. EXPERIMENTAL SETUP

All experiments were conducted using Python-based machine learning libraries, including TensorFlow and Scikit-learn. The models were trained on a standard computing environment with sufficient memory and processing capability to handle image-based data.

Hyperparameters were tuned empirically to achieve optimal performance for each model. The CNN architecture was designed to balance performance and computational efficiency.

Experiments were implemented using TensorFlow (CNN) and Scikit-learn (ML models).

7. LIMITATIONS

Despite promising results, this study has several limitations that must be considered.

First, the dataset used may not be fully representative of global populations. Variations in skin tone, lighting conditions, and image quality can introduce bias, potentially affecting model generalisation across diverse demographic groups.

Second, the CNN model operates largely as a “black-box,” providing limited interpretability. This lack of transparency presents challenges in clinical settings, where understanding the reasoning behind predictions is essential for trust and accountability.

Third, the evaluation was conducted in a controlled experimental setting. Real-world deployment may introduce additional complexities, such as noisy data, inconsistent inputs, and varying environmental conditions.

Fourth, the model does not incorporate uncertainty estimation, which is critical in healthcare scenarios where incorrect predictions may have serious consequences.

Finally, this study does not include clinical validation with medical professionals, which is necessary before practical deployment in healthcare systems.

These limitations highlight the need for further research focusing on fairness, interpretability, robustness, and real-world validation.

8. ETHICAL ANALYSIS & TRUST EVALUATION

While the evaluated models demonstrate promising performance in skin disease classification, their deployment in real-world clinical settings raises significant ethical and socio-technical concerns. This section critically examines these issues through the lenses of bias, explainability, reliability, and regulatory alignment, positioning the system within the broader framework of Responsible AI.

8.1 Dataset Bias and Representation

A key limitation of the dataset used in this study is the potential imbalance in demographic representation, particularly with respect to skin tone diversity. Dermatological conditions often manifest differently across varying skin types, and models trained on datasets dominated by lighter skin tones may fail to generalise effectively to underrepresented populations. This introduces a risk of systematic bias, where certain groups may receive less accurate predictions, potentially leading to inequitable healthcare outcomes.

The absence of explicit demographic metadata further limits the ability to conduct fairness-aware evaluation, making it difficult to quantify disparities in model performance across different population groups. Addressing such biases requires curated datasets, stratified evaluation, and fairness metrics integrated into model validation pipelines.

8.2 Explainability and the Black-Box Problem

Deep learning models, particularly CNNs, are often criticised for their lack of interpretability. In healthcare contexts, where decisions directly impact patient outcomes, the inability to explain model predictions presents a significant barrier to adoption. Clinicians require not only accurate predictions but also insight into the reasoning behind those predictions to ensure alignment with medical knowledge and practice.

The current system operates as a black-box model, providing classification outputs without interpretable justifications. This limits trust and accountability. Incorporating Explainable AI (XAI) techniques such as Grad-CAM or saliency maps could provide visual explanations, enabling clinicians to understand which features of an image contributed to the model's decision.

8.3 Reliability and Clinical Risk

Although the CNN model achieved a validation accuracy of 73.18%, this level of performance may be insufficient for deployment in high-stakes medical environments. Even small error rates can translate into significant clinical consequences, including misdiagnosis or delayed treatment.

Furthermore, the model's performance is evaluated in a controlled experimental setting and may not reflect real-world variability such as image quality, lighting conditions, or patient diversity. Over-reliance on such systems without appropriate safeguards could lead to automation bias, where clinicians overly trust AI outputs without critical evaluation.

To mitigate these risks, AI systems in healthcare should be designed as decision-support tools rather than autonomous decision-makers, ensuring that final judgement remains with qualified medical professionals.

8.4 Human-AI Interaction and Decision Boundaries

The integration of AI into clinical workflows introduces questions regarding the division of responsibility between human practitioners and automated systems. Determining appropriate decision boundaries is essential to ensure safe and effective use.

Human-in-the-loop (HITL) approaches provide a viable solution, where AI-generated predictions are reviewed and validated by clinicians. Such systems can enhance efficiency while maintaining accountability and professional oversight.

Additionally, user interface design and system transparency play a critical role in fostering appropriate levels of trust. Overconfidence in AI outputs must be avoided through clear communication of model uncertainty and limitations.

8.5 Regulatory and Governance Considerations

The deployment of AI systems in healthcare is increasingly governed by regulatory frameworks such as the European Union’s AI Act, which classifies medical AI systems as high-risk applications. These systems must meet strict requirements related to transparency, risk management, data governance, and human oversight.

The current model, in its experimental form, does not yet meet these regulatory standards. Future development must incorporate compliance mechanisms, including documentation of training data, validation processes, and risk mitigation strategies.

Aligning AI systems with such regulatory frameworks is essential not only for legal compliance but also for building public trust and ensuring ethical deployment.

8.6 Summary of Ethical Implications

This analysis highlights that high predictive performance alone is insufficient for deploying AI systems in healthcare. Key challenges include dataset bias, lack of explainability, uncertain reliability, and regulatory constraints. Addressing these issues requires a holistic approach that integrates technical innovation with ethical design principles and governance frameworks.

By situating this work within the context of Responsible AI, this study underscores the importance of developing AI systems that are not only accurate but also fair, transparent, and accountable.

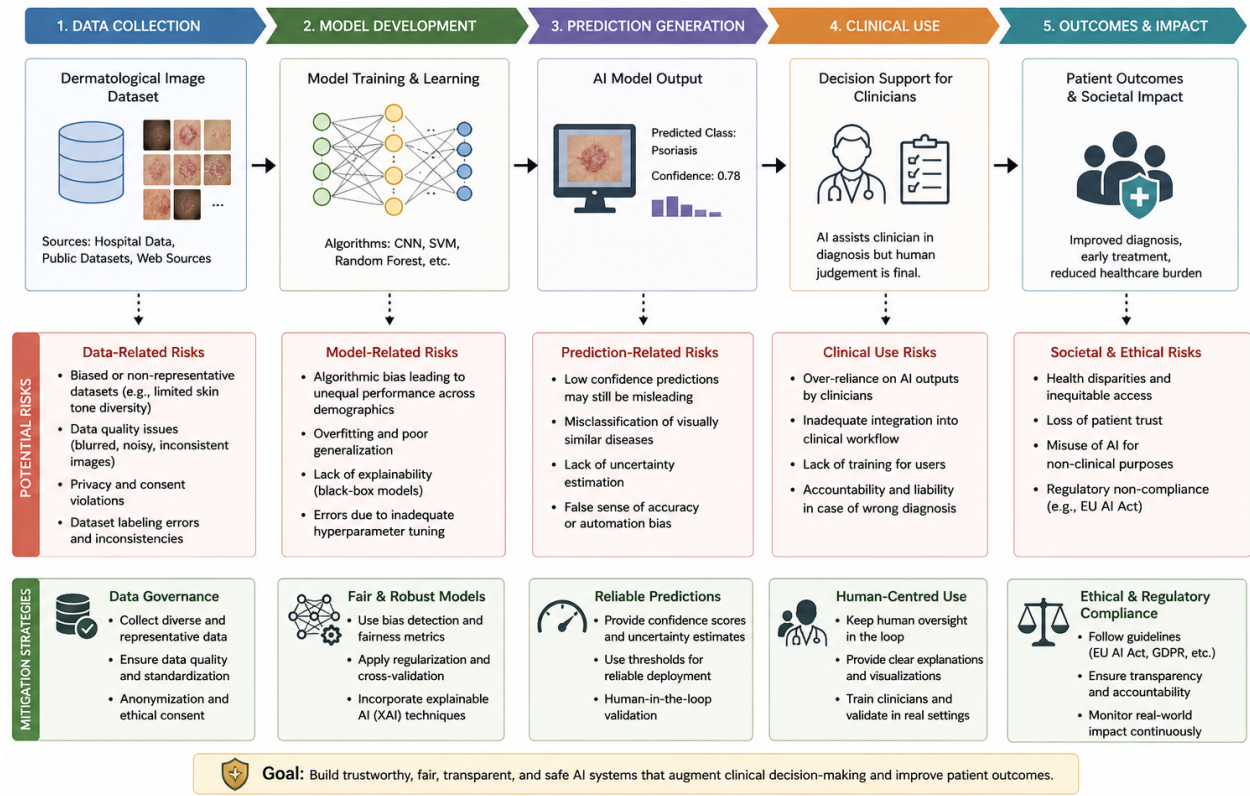


Figure 4: Responsible AI Risk Framework

9. CONCLUSION

This study presented a comparative evaluation of machine learning and deep learning approaches for skin disease classification, demonstrating the effectiveness of Convolutional Neural Networks (CNNs) in achieving superior predictive performance over traditional models. Using a dataset of over 15,000 dermatological images, the CNN model achieved a validation accuracy of 73.18%, highlighting its potential for supporting diagnostic tasks in dermatology.

However, beyond performance metrics, this work emphasised the critical importance of evaluating AI systems through the lens of trustworthiness and ethical responsibility. The analysis revealed several limitations, including potential dataset bias, lack of model interpretability, and concerns regarding reliability in high-stakes clinical environments. These factors underscore that accuracy alone is insufficient when deploying AI systems in healthcare.

By integrating technical evaluation with ethical analysis, this paper contributes to the growing discourse on Responsible AI, particularly in safety-critical domains. It highlights the need for a socio-technical approach that considers not only model performance but also fairness, transparency, and accountability.

Ultimately, this work demonstrates that while AI has the potential to significantly enhance healthcare delivery, its adoption must be guided by rigorous evaluation, ethical design principles, and appropriate governance frameworks to ensure safe and equitable outcomes.

This work highlights that trustworthy AI in healthcare requires a shift from performance-centric evaluation to socio-technical accountability.

10. FUTURE WORK

Building upon the findings of this study, several directions for future research are identified to enhance the trustworthiness and practical applicability of AI systems in dermatology.

First, the integration of Explainable AI (XAI) techniques, such as Grad-CAM and saliency-based visualisations, can provide interpretable insights into model predictions. This would enable clinicians to better understand and validate AI-assisted decisions, thereby improving transparency and trust.

Second, future work should focus on addressing dataset bias through the collection and curation of more diverse and representative dermatological datasets. Incorporating fairness-aware evaluation metrics and conducting subgroup performance analysis across different skin tones and demographic groups will be critical to ensuring equitable model performance.

Third, the incorporation of uncertainty estimation and confidence scoring mechanisms can improve the reliability of AI systems in clinical settings. Models that can communicate uncertainty may help reduce over-reliance and support more informed decision-making by healthcare professionals.

Fourth, exploring human-in-the-loop (HITL) frameworks will be essential for defining effective collaboration between AI systems and clinicians. Such approaches can ensure that AI serves as a

decision-support tool rather than a replacement for expert judgement, maintaining accountability and safety in medical practice.

Finally, future research should investigate the alignment of AI systems with emerging regulatory frameworks, such as the European Union AI Act, focusing on compliance with requirements related to transparency, risk management, and data governance. Bridging the gap between technical development and regulatory implementation will be crucial for the responsible deployment of AI in healthcare.

These directions collectively position this work within the broader agenda of Responsible AI, aiming to develop systems that are not only technically robust but also ethically sound, socially aligned, and clinically reliable.

11. REFERENCES

- [1] Esteva et al., “Dermatologist-level classification...”, *Nature*, 2017
- [2] EU AI Act, European Commission, 2024
- [3] Doshi-Velez & Kim, “Towards Explainable AI”, 2017